

H WMDP Unlearning Details

Trainable layers and parameters We use LoRA (Hu et al., 2021) with rank 64 for GA and GA-LAT. For RMU and RMU-LAT, we do not use LoRA and instead train the MLP weights full-rank, as in Li et al. (2024a).

PGD/RMU layers There are three layer choices that can be varied in our setup: which layer(s) of the model to put the adversary, which layers to train for RMU, and which layer to do the RMU MSE activation matching over. We kept to the same layers (trainable and RMU matching) for RMU as in Li et al. (2024a) – the RMU layer ℓ for the activation matching, with $\ell, \ell - 1, \ell - 2$ trainable to keep the set of hyperparameters to search over reasonably small. Applying attacks to layer $\ell - 2$ requires a smaller ϵ ball radius for our random perturbations; else, we found that the adversary prevents the model trained with RMU from successfully unlearning. We also find the greatest benefit in applying attacks to the layer before the RMU activation matching layer.